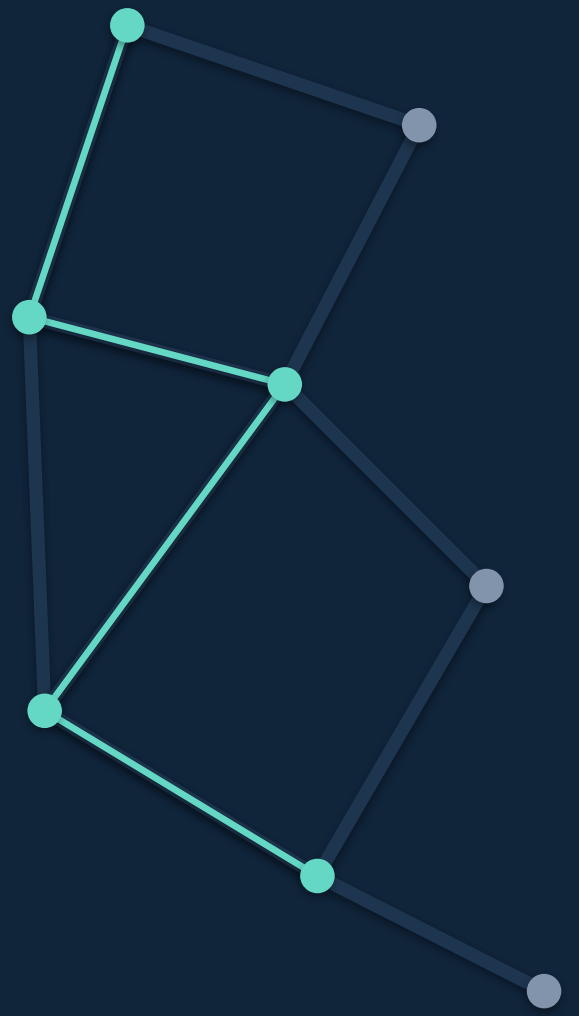
Decoupled Intelligence

A Multi-Agent LLM Framework for Controllable Traffic Scenario Generation in SUMO



INTENT → EXECUTION

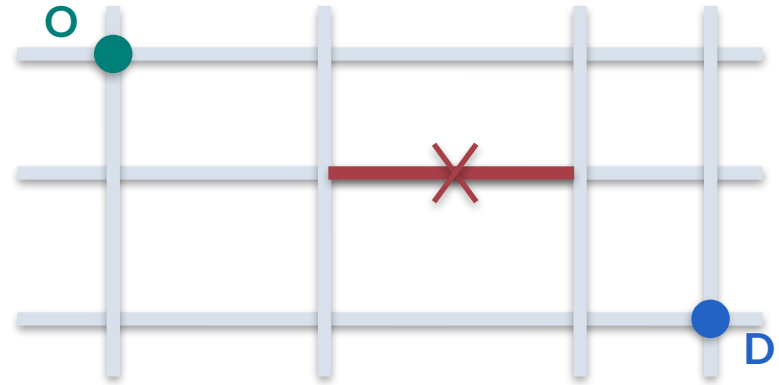
Shuyang Li · Ruimin Ke

Civil and Environmental Engineering
Rensselaer Polytechnic Institute

One traffic question becomes five dependent tasks

ILLUSTRATIVE USER REQUEST

“Close a street in Boston, route 800 vehicles per hour, and report average waiting time.”



Road closure • schematic

01 Build
Road network

02 Modify
Street closure

03 Demand
O–D flows

04 Run
SUMO execution

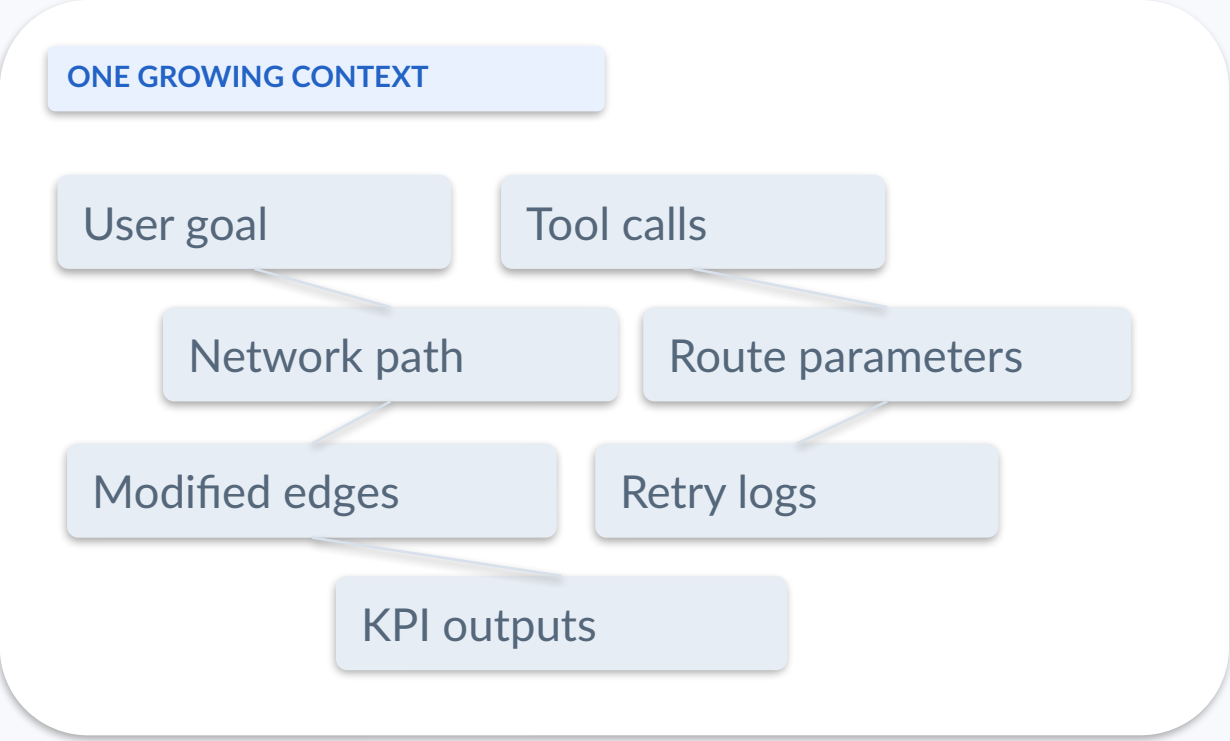
05 Analyze
Waiting time

Every stage must agree on the network, demand, parameters, and files.

Source: Sec. I; Fig. 3 (adapted example)

A single context must remember every dependency

The paper's baseline is a monolithic ReAct agent with all tools and execution history.



- **Path reference errors**

  A downstream step points to the wrong file.

- **Parameter inconsistency**

  Physical values drift across tool calls.

- **Reasoning drift**

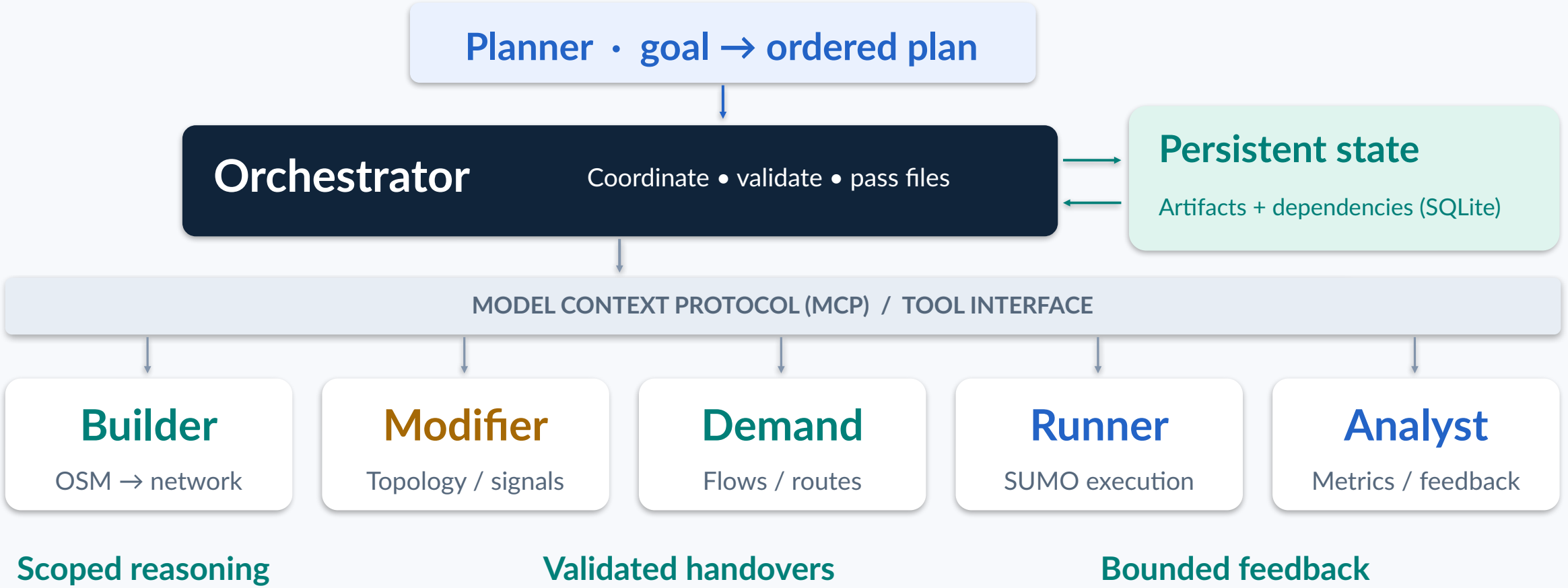
  Retries lose track of the original objective.

Design question: which responsibilities should be explicit in the system?

Source: Sec. IV-A, IV-C, IV-E

Separate reasoning roles. Make state explicit.

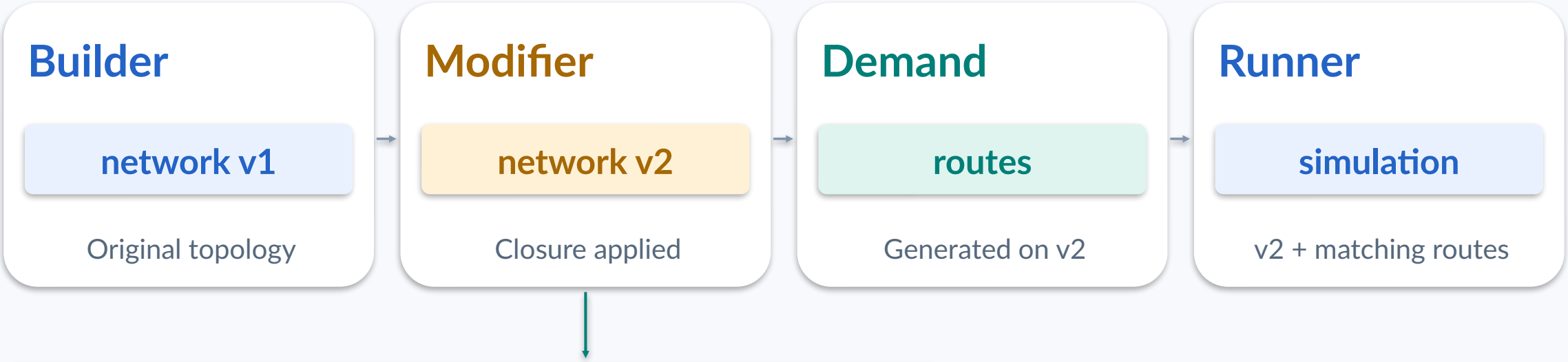
One Planner + five specialized workers, coordinated through a persistent Orchestrator.



Source: Sec. III; Fig. 2 (redrawn)

Route generation uses the validated network

A road closure changes the physical world that downstream tools must use.



Shared state passes the latest validated artifact.

Dependency rule

Modify topology before generating demand.

Source: Sec. III-A, III-B, III-C.2–4 (illustrative artifact labels)

The Planner turns intent into ordered contracts

Illustrative Boston request, adapted from the paper's Fig. 3.

Boston: close Boylston Street; Newbury Street → Beacon Street at 800 veh/h.
Run the simulation and report average waiting time.

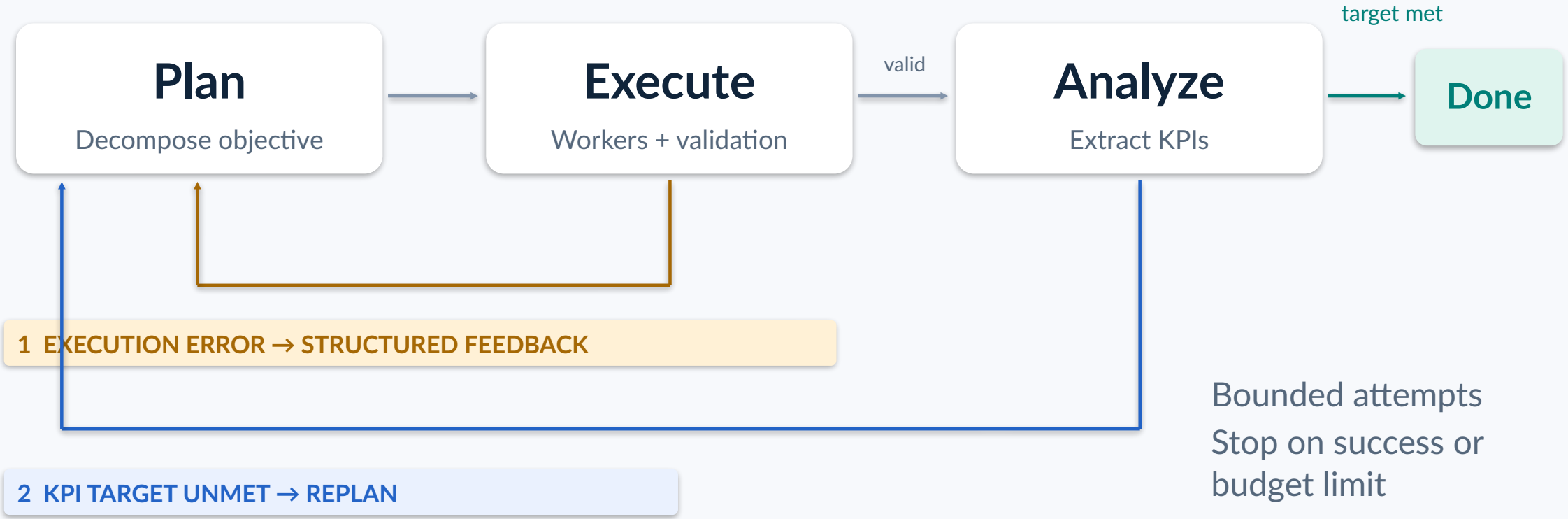
01	**Builder**	Build the Boston road network	.net.xml
02	**Modifier**	Apply the requested Boylston Street closure	Updated network
03	**Demand**	Newbury Street → Beacon Street; 800 veh/h	.rou.xml
04	**Runner**	Configure and execute SUMO	.sumocfg + outputs
05	**Analyst**	Extract average waiting time	KPI report

Each step activates one role and supplies only the parameters that role needs.

Source: Fig. 3; Appendix II–III (simplified, not an execution log)

Two feedback loops address two different problems

Execution repair handles tool failures; KPI feedback revisits the scenario objective.



Bounded attempts

Stop on success or budget limit

Source: Sec. III-B; Algorithm 1

30 tasks test increasing dependency complexity

SUMO 1.21.0 · Qwen3-30B-Coder · monolithic ReAct vs. the proposed framework

L1

Basic

Build a geographic network + qualitative traffic

L2

Intermediate

Specify origin–destination flow + align tool parameters

L3

Complex

Modify network topology before demand generation

WHAT IS MEASURED?

Success rate ↑

Valid configuration + error-free execution

Average tokens ↓

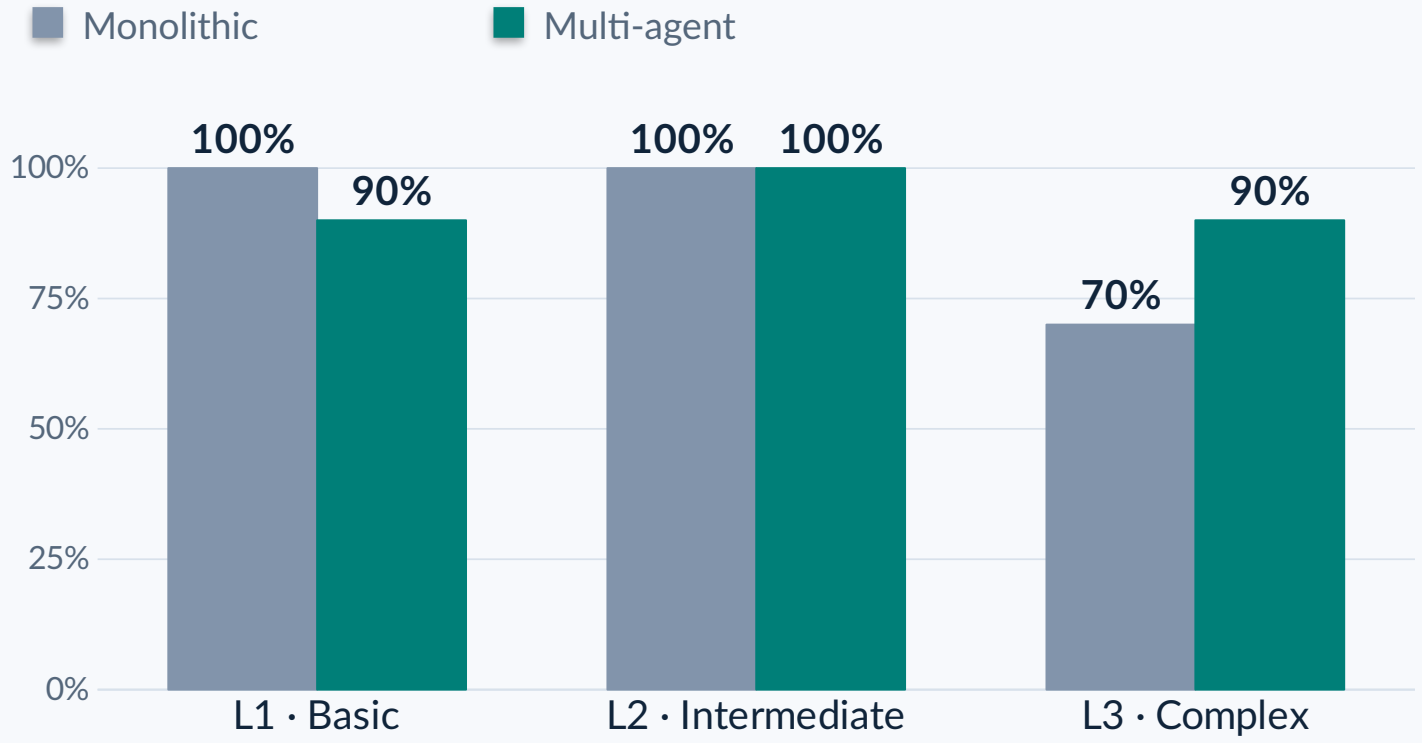
Total LLM input + output tokens

Time-to-Insight ↓

User request → first valid KPI

Complex-task success rises from 70% to 90%

Success rate by task complexity (%) ↑



L3 / COMPLEX

+20

percentage points

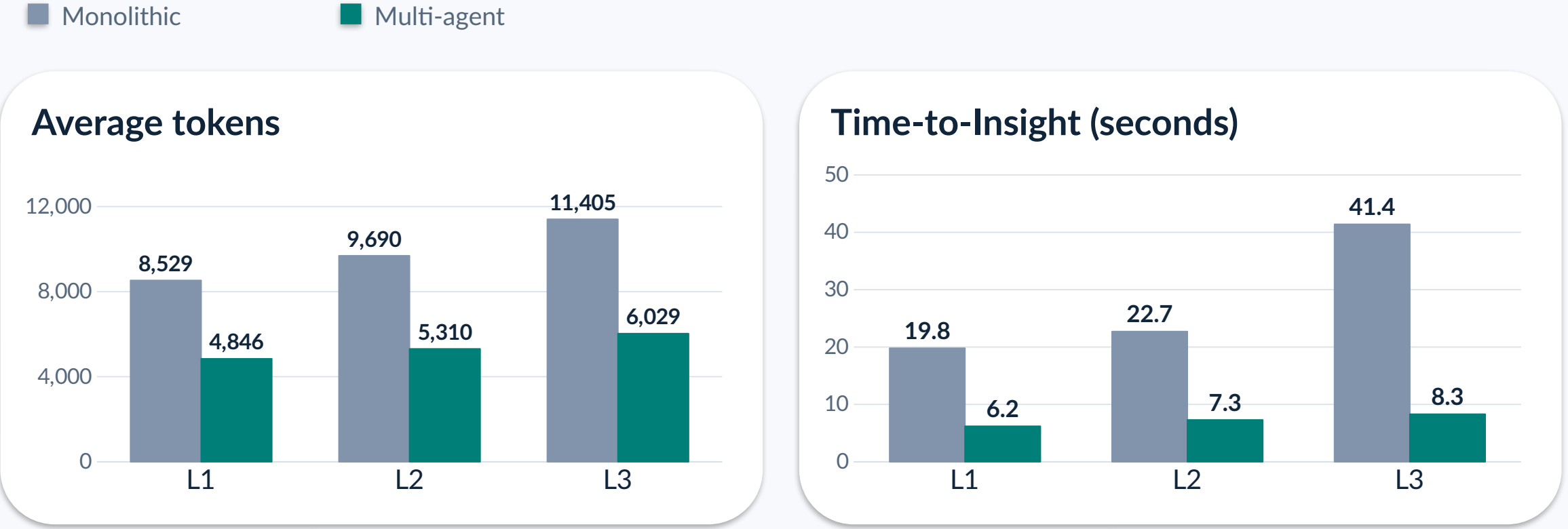
The gain is concentrated in harder workflows.

L1 is 10 points lower; L2 is unchanged. Role separation helps most when dependencies grow.

Source: Table I; Sec. IV-C

Fewer tokens and a shorter path to the first KPI

Reported averages in the architecture comparison; lower is better.



L3: 47.1% fewer tokens

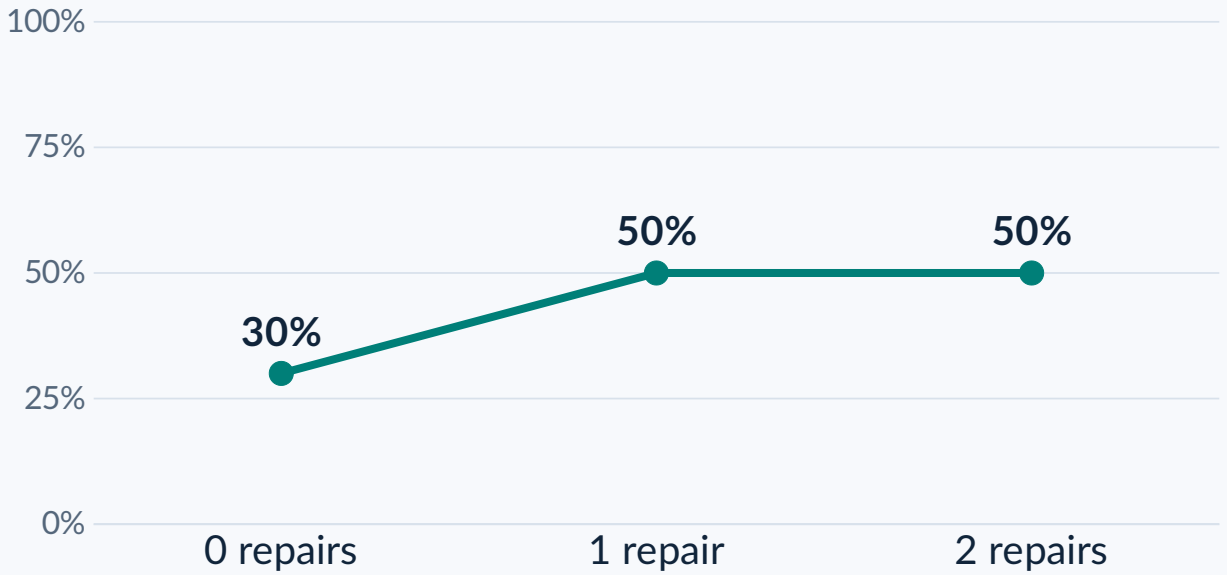
L3: 41.4 s → 8.3 s / ≈5.0× speedup

Source: Table I; percentages and ratios calculated from reported values

One repair attempt helps; a second adds cost

Separate experiment: compare 0, 1, and 2 maximum repair attempts within Table II.

L2 and L3 success rate (%)



L2 and L3 have identical reported success rates.

L3 average tokens

0 repairs	3,981.6
1 repair	6,609.9
2 repairs	9,214.1

More retries ≠ higher success

Simple L1 tasks: 90% → 80% → 90%; unnecessary repair can misread benign warnings.

Source: Table II; Sec. IV-D

What this evidence establishes—and what remains open

A promising workflow result with a bounded empirical scope.

Supported by this study

Higher L3 execution success

Lower tokens and reported TTI

Recovery gains with one repair

Open questions

30 tasks; one evaluated backbone

Components not isolated individually

KPI optimization lacks a quantitative convergence or optimality result

Next: broader benchmarks, component-level ablations, and measured KPI optimization.

Source: Sec. IV–V; reporting limitations identified from the paper

TAKEAWAY

Reliable automation needs explicit coordination

Specialized reasoning + persistent artifacts + bounded feedback

90%

L3 task success

70% monolithic baseline

−47%

L3 token use

11,404.7 → 6,028.7

≈5×

L3 TTI speedup

41.4 s → 8.3 s

Natural-language objectives → executable, inspectable SUMO workflows.

Evidence: Table I, complex tasks (L3)

Thank you

Questions?

Decoupled Intelligence

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Supported by the IBM–RPI Future of Computing Research Collaboration.

Full results: monolithic vs. multi-agent

Exact values from Table I. ↑ Higher is better; ↓ lower is better.

Complexity	Metric	Monolithic	Multi-agent
L1 · Basic	Success rate (%) ↑	100.0	90.0
	Average tokens ↓	8,529.2	4,845.5
	TTI (s) ↓	19.8	6.2
L2 · Intermediate	Success rate (%) ↑	100.0	100.0
	Average tokens ↓	9,690.3	5,310.1
	TTI (s) ↓	22.7	7.3
L3 · Complex	Success rate (%) ↑	70.0	90.0
	Average tokens ↓	11,404.7	6,028.7
	TTI (s) ↓	41.4	8.3

Overall architecture comparison; individual components are not isolated in this table.

Full results: the repair-budget experiment

Exact values from Table II. Compare repair budgets within this experiment.

Complexity	Metric	0 repairs	1 repair	2 repairs
L1 · Basic	Success rate (%) ↑	90.0	80.0	90.0
	Average tokens ↓	4,969.1	5,530.3	6,752.6
L2 · Intermediate	Success rate (%) ↑	30.0	50.0	50.0
	Average tokens ↓	3,601.6	6,829.5	8,484.8
L3 · Complex	Success rate (%) ↑	30.0	50.0	50.0
	Average tokens ↓	3,981.6	6,609.9	9,214.1

No additional L2/L3 success gain from a second attempt; token use continues to rise.

Source: Table II; Sec. IV-D

What each worker receives and produces

Role contracts and artifact dependencies, simplified from the methodology and prompt appendices.

Worker	Required context	Responsibility	Output
Builder	City / geographic extent	Build network	.net.xml
Modifier	Network + requested edit	Apply topology / TLS edit	Updated .net.xml
Demand	Validated network + flow / O–D	Generate matching demand	.rou.xml
Runner	Network + routes + run settings	Configure and execute	.sumocfg + outputs
Analyst	Simulation outputs + metric	Aggregate KPIs	Metrics + feedback

One active role per planned step

Skip Modifier when no edit is requested

Source: Sec. III-C; Appendix II–III

Answers to likely design questions

Why not a stronger single agent?	This study compares one backbone and one monolithic ReAct design. Broader baselines remain to be tested.
Does MCP itself prevent path errors?	MCP standardizes tool access. The Orchestrator manages validated artifact references and dependencies.
Is traffic KPI optimization proven?	The feedback mechanism is designed for it. Reported quantitative results focus on execution and repair.
Do more agents or retries always help?	No. L1 success is lower in Table I; a second repair adds tokens without improving L2/L3 success in Table II.